

KARRI JYOTSHNA RAMA LAKSHMI

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Summary

ECE student (B.Tech, CGPA 8.78, 2027 batch) with a strong foundation in RTL design (Verilog), embedded systems (C/ESP32), and self-directed learning. Passionate about CPU microarchitecture, performance analysis, and hardware-software integration. Currently in a 100-day VLSI journey, building and verifying modules daily.

Technical Skills

- **RTL & Digital Design:** Verilog, Finite State Machines, Testbenches, Simulation (Icarus Verilog, GTKWave)
- **CPU Architecture Basics:** 5-stage pipeline, hazards, forwarding, branch prediction (conceptual), cache hierarchy, out-of-order execution (basic), x86 ISA (registers, memory operations)
- **Performance Metrics:** IPC, CPI, latency, throughput – measured simulation cycles for RTL modules
- **Embedded Systems:** C, Embedded C, ESP32, Arduino, UART, I2C, SPI, Bluetooth
- **Scripting & Automation:** Git, GitHub, VS Code
- **AI/ML Interest:** Built AI agent system (SVASTI) – interested in AI-driven performance automation

Projects & RTL Work

SVASTI – AI Traffic Prediction Platform (HTML/CSS/JS)

- Multi-agent system (Predictor, Optimizer, Nudge, Verifier) for real-time congestion forecasting.
- Semi-Finalist at ET AI Hackathon 2026 (top 12% of 54,000+ participants). Demonstrated system integration and simulation.

MotoScore – Riding Behavior Analyzer (ESP32, MPU6050, GPS)

- Real-time firmware reading MPU6050 (I2C) and GPS (UART). Bench-tested sensor fusion, reduced harsh events by 30%.
- Demonstrated ability to capture and analyse performance data.

RTL Modules (Verilog) – 100-day VLSI Journey

- Designed and simulated UART transmitter, synchronous FIFO, 4-bit ALU, sequence detectors (FSM), counters, shift registers.
- Wrote self-checking testbenches, generated waveforms with GTKWave. Focused on cycle-accurate simulation and performance trade-offs.

Low-Cost Smoke Detector & Paddy Husking Machine (Arduino, C++)

- Prototype circuit design, actuator control, hardware validation – demonstrated systematic debugging.

Internships & Training

Trainee (6 months) AP TRANSCO, Visakhapatnam | 2023–2024

- Power distribution monitoring, telecom role in grid systems – exposure to industrial electronics.

Shortlisted Candidate (NATS) Hindustan Shipyard Ltd (HSL), DAT role | Feb 2026

- 1-week orientation in industrial electronics.

APSSDC Virtual Internship Dual Axis Solar Tracker (Arduino, LDRs, servos)

Achievements

- Winner – AU Hackathon (PregoCare)
- Top 11% (6k/54k) – ET AI Hackathon Urban Transportation
- 100-day VLSI journey (ongoing, publicly documented on GitHub)

Education

B.Tech – Electronics & Communication Engineering NSRIT | 2024–2027 | CGPA: 8.78

Diploma – Electronics & Communication Engineering Govt Polytechnic for Women, Bheemunipatnam | 2021–2024 | 84.5%

Soft Skills & Interests

- Strong analytical and problem-solving skills
- Quick learner – independent study of RTL, CPU architecture, and performance metrics
- Interest in AI/ML for hardware performance automation
- Good written and verbal communication (English)